

[DC] Improving XR Training and Collaboration Through Shared, Self, and Mediated Awareness

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ABSTRACT

Extended Reality (XR) creates new possibilities for training and collaboration by merging digital information with the physical space. However, sustaining *awareness*, the ongoing understanding of what others (or oneself) are doing, seeing, and needing, remains challenging. My PhD explores how awareness emerges in XR training spaces, and how it can be maintained between people, within oneself, and in interaction with intelligent agents.

Index Terms: Extended Reality, Awareness, Training, Collaboration, Adaptive Feedback, Distributed Perspectives.

1 INTRODUCTION

Extended Reality (XR) has the potential to change how we connect and collaborate. Particularly for training scenarios, XR shows promise in increasing the scale and efficiency of how we learn and teach new skills. However, while XR can offer richer spatial and embodied cues [8], many existing systems remain one-off solutions that do not meaningfully support scalable skill acquisition. Sustaining *awareness*, the continuous perception of what others or oneself are doing, seeing, and needing [5], remains challenging, especially when perspectives differ, feedback is delayed, or collaboration involves multiple roles.

To better understand these challenges, we conducted a **systematic survey of XR training systems** [3], where we analyzed collaboration structures, perspective symmetry, feedback modality, and temporal coupling, to understand how existing approaches support awareness between experts and trainees. This analysis revealed key gaps: (i) limited support beyond one-expert-to-one-trainee set-ups; (ii) few systems enabling mixed perspectives and transitions across different immersion degrees; and (iii) over-reliance on human experts and synchronous interaction for feedback.

To address these gaps, my research investigates how we can leverage different mechanisms and dynamics to support awareness across three spheres: *between people*, *within oneself*, and *between humans and agents* (Figure 1). Specifically, my research explores the following research questions:

- **RQ1.** How do perspective sharing mechanisms impact learning, coordination, and error recovery in XR?
- **RQ2.** How do feedback mechanisms impact self-regulated learning, coordination, and error recovery in XR?
- **RQ3.** How do these scale from dyads to small groups?
- **RQ4.** What changes with automated guidance?

2 SHARED AWARENESS: BETWEEN USERS

In collaborative XR environments, shared awareness depends on whether users can interpret each other's actions despite spatial and perceptual differences [13], particularly when collaborators gesture at 3D objects, which can be occluded or seen from different

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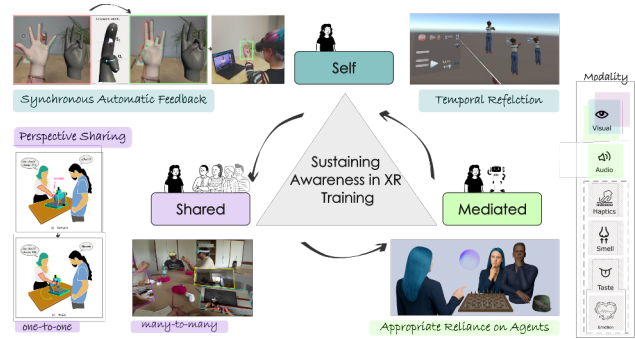


Figure 1: Sustaining awareness in XR training through shared, self, and mediated awareness.

views [11]. Hence, I investigate how XR can reduce perceptual gaps between collaborators by aligning viewpoints, revealing occluded information, and supporting embodied communication cues.

In **MAGIC** [2], we address these challenges by reinterpreting pointing gestures across coordinate spaces to maintain referential intent between collaborators. We distorted the remote user's gestures to correctly reflect them in the local user's space, enabling users to *share the same point of view of the 3D space while face-to-face*, maintaining non-verbal communicational cues' advantages [1]. Results from a user study with 12 participants suggest that, in face-to-face collaboration settings, MAGIC significantly improves *pointing agreement*, co-presence and awareness of interactions performed in the shared space, increasing task performance.

Other works have explored perspective sharing beyond dyads [6], but most provide only limited bi-directional sharing. I am interested in how enabling multi-directional perspective sharing could improve hands-on training in small groups. My ongoing work **PeeRXpective**, explores how we can enable shared awareness in multi-user settings through omni-directional perspective sharing: from expert to trainees, trainees to expert, and between trainees. We introduce *opportunistic multi-perspective sharing*, and explore how accessing everyone's first-person view dynamically positioned in space as needed can improve training. Specifically, we are in the process of running a user study with 60 participants across 12 groups of 1 expert and 4 trainees, in a crochet-based task, aiming to understand how this setup influences group coordination, awareness, and learning; and how it might impact expert's workload.

3 SELF AWARENESS: UNDERSTANDING ONE'S ACTIONS

Awareness in training is not limited to interaction: people must also interpret and adapt their own actions, especially when no expert is available. In this second line of work, I explore how automated feedback mechanisms can assist self-awareness in hands-on tasks.

In **Hand Augmentations** [4], we examined how visual feedback mechanisms can help users self-correct gestures without a live instructor, by providing *real-time corrective feedback through dynamic hand augmentations* that change color, shape, and placement. We varied feedback style (color vs. shape) and spatial placement (superimposed vs. adjacent) to assess how automated visual

feedback affected preference, performance and short term recall. Insights from a user study with 18 participants showed color-based and superimposed feedback improved task efficiency, while adjacent cues were preferred for comfort, highlighting trade-offs between efficiency and user ease.

We are extending this idea of visual self-correction across time, exploring *parallel embodiment* [12] for productivity and self-guidance. In our ongoing work, **VR Duplicates**, users can record and replay their own embodied actions as multiple “*VR ghosts*”, allowing reflection on past performances and planning of future ones. This adds a temporal dimension to self-awareness, which we hypothesize can help users monitor progress and refine skills through reflection and analysis beyond the moment of interaction.

4 MEDIATED AWARENESS: BETWEEN HUMANS & AGENTS

As AI becomes increasingly integrated into training, a new form of awareness emerges: between humans and intelligent systems. Here, the challenge lies not only in feedback accuracy but in calibrating *appropriate reliance* ([10]): when to trust and when to question automated guidance. Specifically, I question if users’ reliance on these agents is shaped by surface-level cues such as appearance and perceived expertise.

Our ongoing work on **AI Agents Perception** investigates how agent appearance influences decision-making in high-stakes training tasks by possibly affecting user’s awareness of their own capabilities. We are currently preparing a VR user study where participants will collaborate with an AI agent with differently styled representations (varying level of anthropomorphism, expertise cues, and user similarity) in the context of a chess play. We hypothesize that representational cues shape how users perceive agent reliability and their own role in the interaction, and we aim to understand how awareness of AI agents’ capabilities might be shaped by simple features such as appearance, and how we can design agents that appropriately convey these capabilities while maintaining user’s understanding and trust.

5 EXTENDING AWARENESS: EXPLORING EMOTIONAL, AND MULTISENSORY DIMENSIONS

Looking forward, I plan to explore some complementary directions related to how awareness is shaped by users’ emotional states, and how to leverage multisensory feedback to enhance it.

Adapting Interactions Based on Users’ Affect. Emotions strongly shape how people interpret feedback, coordinate with others, and engage with a task[9]. However, XR systems rarely adapt interactions based on how users feel [7]. I plan to explore how we can adapt afforded interactions to user’s emotional cues, determining how and when emotions should be supported, reinforced, or gently mitigated to sustain productive engagement. I want to design and implement different interaction styles that allow users to amplify or counteract their emotional states, and conduct a user study to understand how awareness of the affective experience can shape interactions in the XR space.

Exploratory Multisensory Feedback. Beyond visual cues, other sensory channels such as haptic, auditory, or proprioceptive feedback may provide complementary information, compensating for limitations of visual-only feedback and enhancing awareness across different spheres. Aligning feedback across multiple sensory modalities may increase the clarity and salience of feedback cues, making them easier for users to interpret and respond to. As an example, during an assembly task, users might feel vibrations signaling hand-placement errors while hearing auditory cues indicating task progress. I want to explore how combining multisensory feedback with adaptive XR systems affects cognitive load, engagement, and performance.

6 CONCLUSION

Current approaches to XR training offer limited support for users to efficiently learn and communicate skills. My research aims to demonstrate how different mechanisms related to enabling shared references, provide feedback and appropriately rely on information can support awareness across three spheres: between people, within oneself, and between humans and agents. I envision *awareness* as a guiding principle for XR systems, shaping not only what users see but also how they understand, respond, and connect with others, themselves, and the intelligent systems that are increasingly becoming part of their environments. By validating this framework across multi-user collaboration, self-guided learning, and intelligent-agent scenarios, I aim to contribute both actionable design principles and empirically grounded insights for developing adaptive XR systems that sustain awareness in dynamic training environments.

REFERENCES

- [1] S. Benford, J. Bowers, L. E. Fahlén, C. Greenhalgh, and D. Snowdon. User embodiment in collaborative virtual environments. *ACM CHI*, 1995. doi: 10.1145/223904.223935 1
- [2] C. G. Fidalgo, M. Sousa, D. Mendes, R. K. Dos Anjos, D. Medeiros, K. Singh, and J. Jorge. Magic: Manipulating avatars and gestures to improve remote collaboration. *IEEE VR*, 2023. doi: 10.1109/VR55154.2023.00059 1
- [3] C. G. Fidalgo, Y. Yan, H. Cho, M. Sousa, D. Lindlbauer, and J. Jorge. A survey on remote assistance and training in mixed reality environments. *IEEE TVCG*, 2023. doi: 10.1109/TVCG.2023.3247081 1
- [4] C. Gonçalves Fidalgo, Y. Yan, M. Sousa, J. Jorge, and D. Lindlbauer. Exploring ar hand augmentations as error feedback mechanisms for enhancing gesture-based tutorials. *Frontiers in VR*, 2025. doi: 10.3389/frvir.2025.1574965 1
- [5] C. Gutwin and S. Greenberg. A descriptive framework of workspace awareness for real-time groupware. *ACM CSCW*, 2002. doi: 10.1023/a:1021271517844 1
- [6] A. H. Hoppe, F. van de Camp, and R. Stiefelhagen. Shisha: Enabling shared perspective with face-to-face collaboration using redirected avatars in virtual reality. *ACM CSCW*, 2021. doi: 10.1145/3432950 1
- [7] G. Lampropoulos, P. Fernández-Arias, Á. Antón-Sancho, and D. Vergara. Affective computing in augmented reality, virtual reality, and immersive learning environments. *Electronics*, 13(15):2917, 2024. 2
- [8] S. Orts-Escolano, C. Rhemann, S. Fanello, W. Chang, A. Kowdle, Y. Degtyarev, D. Kim, P. L. Davidson, S. Khamis, M. Dou, V. Tankovich, C. Loop, Q. Cai, P. A. Chou, S. Mennicken, J. Valentin, V. Pradeep, S. Wang, S. B. Kang, P. Kohli, Y. Lutchyn, C. Keskin, and S. Izadi. Holoportation: Virtual 3d teleportation in real-time. *ACM UIST*, 2016. doi: 10.1145/2984511.2984517 1
- [9] R. W. Picard. *Affective computing*. MIT press, 2000. 2
- [10] M. Schemmer, P. Hemmer, N. Kühl, C. Benz, and G. Satzger. Should i follow ai-based advice? measuring appropriate reliance in human-ai decision-making. *arXiv preprint arXiv:2204.06916*, 2022. 2
- [11] M. Sousa, D. Mendes, R. K. d. Anjos, D. S. Lopes, and J. Jorge. Negative space: Workspace awareness in 3d face-to-face remote collaboration. In *Proceedings of the 17th ACM SIGGRAPH International Conference on Virtual-Reality Continuum and its Applications in Industry*, pp. 1–2, 2019. 1
- [12] K. Takada, M. Kawaguchi, A. Uehara, Y. Nakanishi, M. Armstrong, A. Verhulst, K. Minamizawa, and S. Kasahara. Parallel ping-pong: Exploring parallel embodiment through multiple bodies by a single user. *ACM AHs*, 2022. doi: 10.1145/3519391.3519408 2
- [13] S. Whittaker. Things to talk about when talking about things. *Human-Computer Interaction*, 18(1-2):149–170, 2003. 1